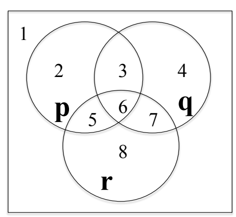
Assignment 3 - PHP, Propositional Logic and Inference laws

Due: November 5 at 10pm NO LATE ASSIGNMENTS ACCEPTED.

Submit your assignment by the due date to Gradescope. Recall that the rules of academic integrity for this class allow you to discuss the questions with your peers, but the write up must be your own. You ***may not use any other sources*** *such as the internet or other textbooks to find solutions to these questions. Your answers will be graded both on their final solution as well as the work you show to achieve that solution. Note that a subset of the questions will be graded. By submitting your assignment you agree to abide by these rules.*

1. Given 10 numbers $a_1, a_2, \ldots, a_{10}$, from the set S of natural numbers $S = \{1, 2, 3, \ldots, 100\}$. Prove that there exist two (or more) subsets of the 10 numbers whose elements sum to the same value. Food for thought - how would you determine that 10 values are needed?
2. Given the set of S natural numbers $S = \{1, 2, 3, \ldots, 100\}$ select fifty one numbers from S. Prove that at least one of the numbers you chose is a multiple of another number that you chose.
3. Evan is training for baseball season. His coach wants him to pitch or hit 90 times over the course of the **53 days** before the season starts. He plans to do at least one pitching or hitting session a day. Prove that there exists a span of consecutive days that Evan will complete exactly 15 sessions.
4. Use the following Venn diagram for this question.



   (a) List the regions of the Venn diagram that make the following statement true: $(\neg p \to q) \to r$.

   (b) Construct an equivalent statement to the one in (a) using only $\wedge$, $\vee$, and/or $\neg$ using as few connectives as possible.

   (c) List the regions of the Venn diagram that make the following statement true: $p \to (q \to \neg r)$.

   (d) Is $\to$ commutative? associative? Explain your answer.

5. A technician suspects that one or more processors in a distributed system is not working properly. The processors, A, B, and C, are all capable of reporting information about the status of the processors in the system (either working or not working). The technician is unsure whether a processor is really not working, or whether the problem is in the status reporting routines in one or more of the processors. After polling each processor, the technician receives the following status reports: [10]

   - Processor A reports that Processor B is not working and Processor C is working.
   - Processor B reports that Processor A is working if and only if Processor B is working.
   - Processor C reports that at least one of the other two processors is not working.

   (a) Let a = "A is working," b = "B is working," and c = "C is working." Write the three status reports in terms of a, b, and c, using the symbols of formal logic.

   (b) Assume that all of the status reports are true, which processor(s) is/are working?

(c) Assuming that all of the processors are working, which status report(s) is/are false?

(d) Assuming that a processor's status report is true if and only if the processor is working, what is the status of each processor?

6. Prove or disprove the following using equivalence laws. :

   (a) $p \rightarrow (q \rightarrow r) \iff (p \wedge \neg r) \rightarrow \neg q$

   (b) $\neg((p \leftrightarrow q) \vee (p \wedge \neg q)) \iff ((p \leftrightarrow \neg q) \wedge (\neg p \vee q))$

7. A gang's "Boss" is killed by one of it's own members. A detective interviews the four members and determined that all were lying except for one. He deduced who killed the Boss on the basis of the following statements:

   (a) Knuckles: Chuckles killed the Boss.

   (b) Fierce: Grimace didn't kill the Boss.

   (c) Chuckles: Grimace was shooting craps with Knuckles when the Boss was knocked off.

   (d) Grimace: Chuckles didn't kill the Boss.

   Who killed the Boss? Justify your answer.

8. For the following questions, use the inference laws given in class to show that the given conclusion can be logically inferred from the premises.

   (a)

Statement	Reason
$\neg p \rightarrow r \wedge \neg s$	given
$t \rightarrow s$	given
$u \rightarrow \neg p$	given
$\neg v$	given
$u \vee v$	given
$\vdots$	$\vdots$
$\neg t$	conclusion

   (b)

Statement	Reason
$p \rightarrow q$	given
$\neg r \vee s$	given
$p \vee r$	given
$\vdots$	$\vdots$
$\neg q \rightarrow s$	conclusion